ExamGOAL
PYQ PDF

Free · With answer key

GATE ECE · NETWORK THEORY · CHAPTER-WISE PYQS

Sinusoidal Steady State Response

GATE ECE Network Theory · Previous Year Questions with Answer Key

75	39	1987 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	28 questions
Marks 2	38 questions
Marks 5	8 questions
Marks 8	1 question
Answer key	at the end

Practise online

Every question with its full solution, free on **questions.examside.com**

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

Marks 1 28 questions

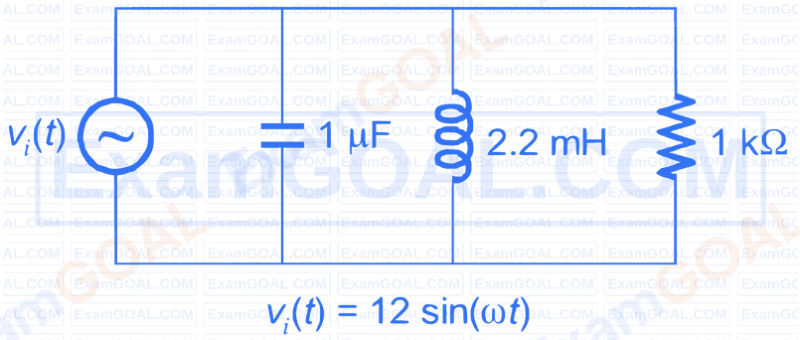
Q1. GATE ECE 2026

Consider the circuit shown in the Figure, where the input $v_i(t)$ is in Volt.

The average power (in mW) dissipated in the load resistance of $1\text{k}\Omega$ at the resonant frequency is ____ .

(rounded off to two decimal places)

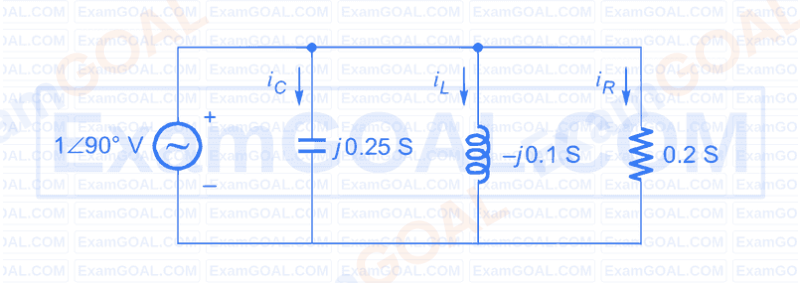
Open solution



Q2. GATE ECE 2025

Let i_C, i_L, and i_R be the currents flowing through the capacitor, inductor, and resistor, respectively, in the circuit given below. The AC admittances are given in Siemens(S). Which one of the following is true?

Open solution



(A) $i_C = 0.25\angle 180^\circ \text{A}, \quad i_L = 0.1\angle 0^\circ \text{A}, \quad i_R = 0.2\angle 90^\circ \text{A}$

(B) $i_C = 4\angle 180^\circ \text{A}, \quad i_L = 10\angle 0^\circ \text{A}, \quad i_R = 5\angle 90^\circ \text{A}$

(C) $i_C = 0.25\angle 270^\circ \text{A}, \quad i_L = 0.1\angle 90^\circ \text{A}, \quad i_R = 0.2\angle 90^\circ \text{A}$

(D) $i_C = 4\angle 90^\circ \text{A}, \quad i_L = 10\angle 270^\circ \text{A}, \quad i_R = 5\angle 0^\circ \text{A}$

Q3. GATE ECE 2023

A series RLC circuit has a quality factor Q of 1000 at a center frequency of 10^6 rad/s. The possible values of R, L and C are

Open solution

(A) $R = 1\,\Omega, L = 1\,\mu H$ and $C = 1\,\mu F$

(B) $R = 0.1\,\Omega, L = 1\,\mu H$ and $C = 1\,\mu F$

(C) $R = 0.01\,\Omega, L = 1\,\mu H$ and $C = 1\,\mu F$

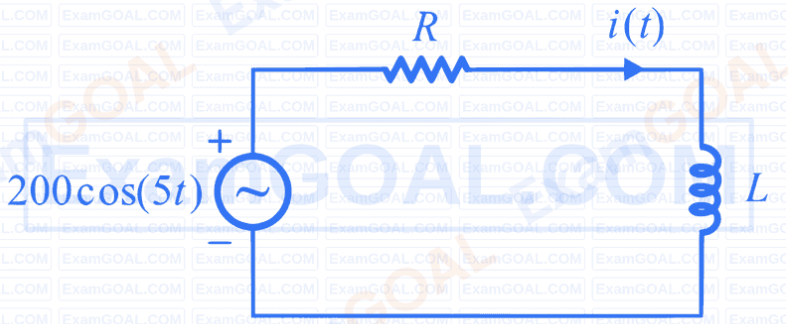
(D) $R = 0.001\,\Omega, L = 1\,\mu H$ and $C = 1\,\mu F$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q4. GATE ECE 2020

The current in the RL - circuit shown below is $i(t) = 10\cos(5t - \pi/4)\text{A}$. The value of the inductor (rounded off to two decimal places) is ____ H.

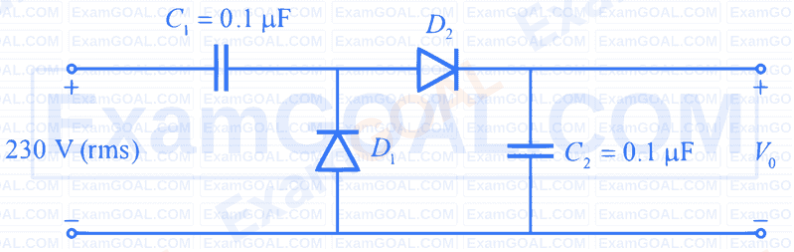
Open solution



Q5. GATE ECE 2020

In the circuit shown below, all the components are ideal and the input voltage is sinusoidal. The magnitude of the steady - state output V_0 (rounded off to two decimal places) is ____ V.

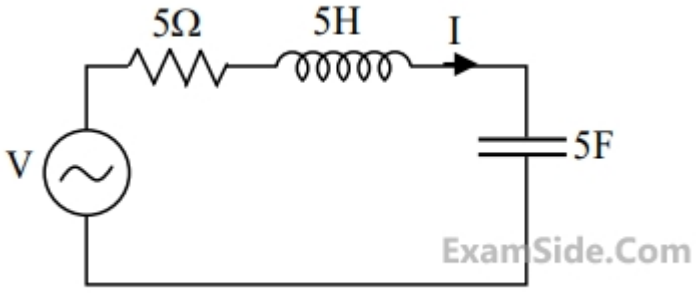
Open solution



Q6. GATE ECE 2017 Set 2

In the circuit shown, V is a sinusoidal voltage source. The current I is in phase with voltage V. The ratio $\frac{\text{Amplitude of voltage across the capacitor}}{\text{Amplitude of voltage across the resistor}}$ is _____.

Open solution

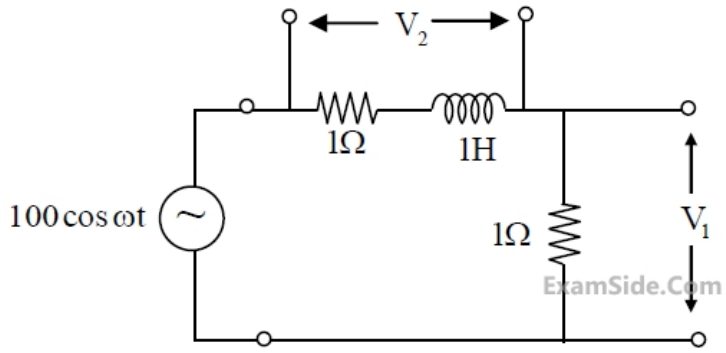


Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q7. GATE ECE 2017 Set 1

Open solution

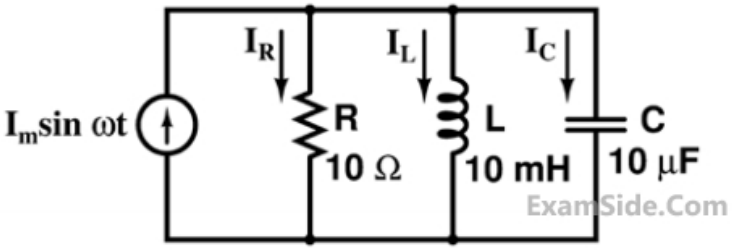
In the circuit shown, the positive angular frequency ω (in radians per second) at which magnitude of the phase difference between the voltages V_1 and V_2 equals $\frac{\pi}{4}$ radians, is _____.



Q8. GATE ECE 2016 Set 2

Open solution

The figure shows an RLC circuit with a sinusoidal current source.



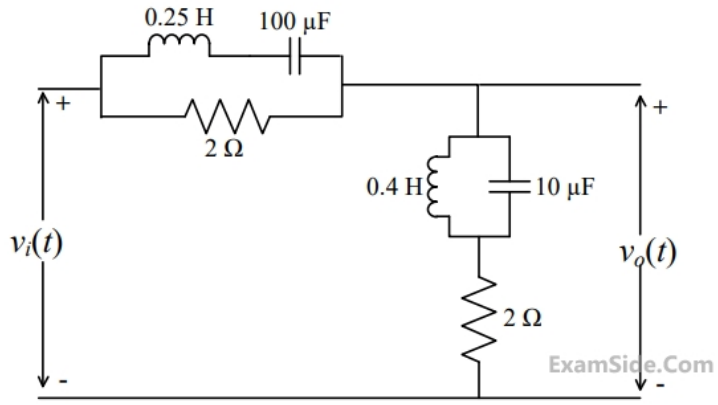
At resonance, the ratio $|\mathrm{I}_L|/\,|\mathrm{I}_R|$, i.e., the ratio of the magnitudes of the inductor current phasor and the resistor current phasor, is _____.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q9. GATE ECE 2016 Set 3

Open solution

In the RLC circuit shown in the figure, the input voltage is given by $v_i(t) = 2\cos(200t)+4\sin(500t)$. The output voltage $v_0(t)$ is



(A) cos(200t) + 2 sin(500t)

(B) 2cos(200t) + 4 sin(500t)

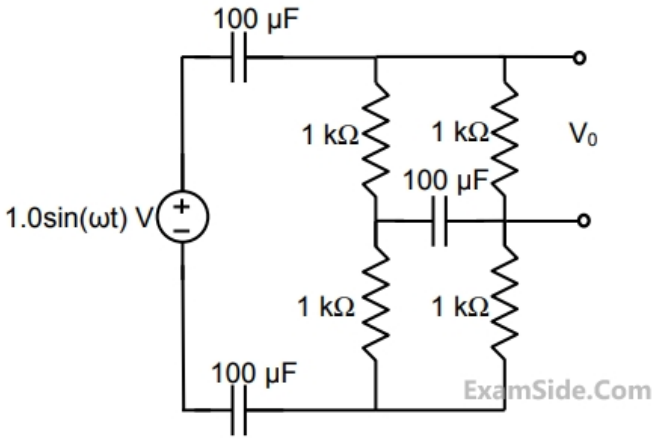
(C) sin(200t) + 2 cos(500t)

(D) 2sin(200t) + 4 cos(500t)

Q10. GATE ECE 2015 Set 3

Open solution

At very high frequencies, the peak output voltage V_0 (in Volts) is _____.

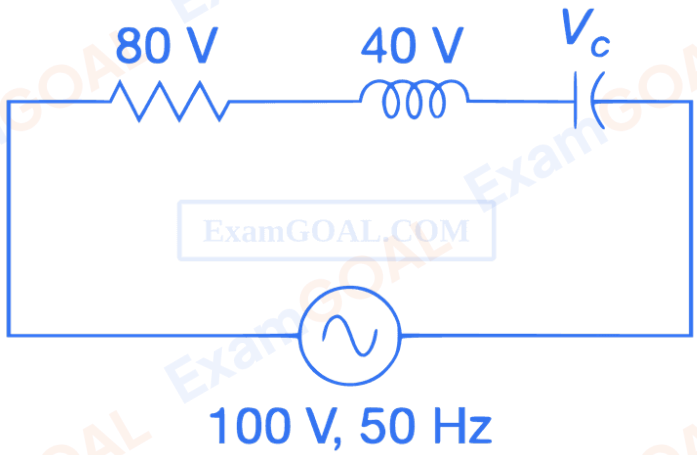


Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q11. GATE ECE 2015 Set 2

The voltage (V_C) across the capacitor (in Volts) In the network shown is _____.

Open solution



Q12. GATE ECE 2015 Set 1

In the circuit shown, at resonance, the amplitude of the sinusoidal voltage (in Volts) across the capacitor is _____.

Open solution



Q13. GATE ECE 2010

For parallel RLC circuit, which one of the following statements is NOT correct?

(A) The bandwidth of the circuit deceases if R is increased

(B) The bandwidth of the circuit remains same if L is increased

(C) At resonance, input impedance is a real quantity

(D) At resonance, the magnitude of input impedance attains its minimum value.

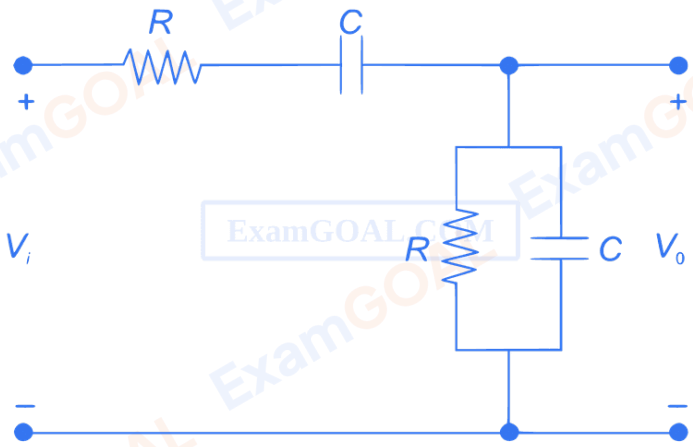
Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q14. GATE ECE 2007

The RC circuit shown in the figure is

Open solution



(A) a low-pass filter
(B) a high-pass filter
(C) a band-pass filter
(D) a band-reject filter

Q15. GATE ECE 2005

In a series RLC circuit $R = 2\,k\Omega,\ \ L = 1H$ and $C = 1/400\,\mu F$. The resonant frequency is

(A) $2 \times 10^4\ Hz$ (B) $\frac{1}{\pi} \times 10^4\ Hz$ (C) $10^4\ Hz$ (D) $2\pi \times 10^4\ Hz$

Open solution

Q16. GATE ECE 2004

The circuit shown in figure, with $R = 1/3\Omega, L = 1/4H, C = 3F$ has input voltage $v\,(t) = \sin\,2t$. The resulting current $i(t)$ is

Open solution

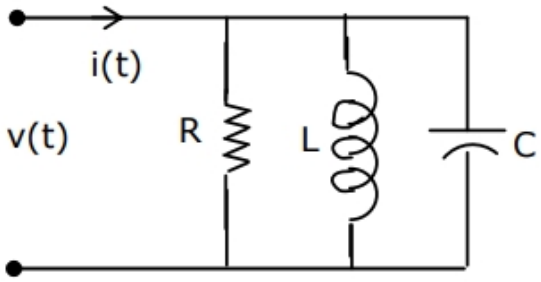


Fig.

ExamSide.Com

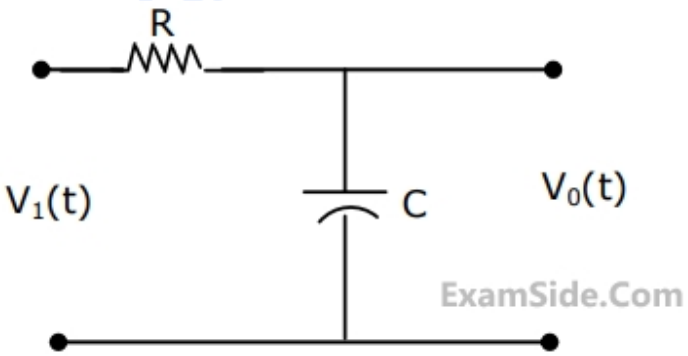
(A) $5\,\sin\left(2t + 53.1^0\right)$
(B) $5\,\sin\left(2t - 53.1^0\right)$
(C) $25\,\sin\left(2t + 53.1^0\right)$
(D) $25\,\sin\left(2t - 53.1^0\right)$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q17. GATE ECE 2004

Open solution

For the circuit shown in figure, the time constant $RC = 1\ ms$. The input voltage is $v_i(t) = \sqrt{2}\ \sin\ 10^3 t$. The output voltage $v_0(t)$ is equal to



(A) $\sin\left(10^3 t - 45^0\right)$
(B) $\sin\left(10^3 t + 45^0\right)$
(C) $\sin\left(10^3 t - 53^0\right)$
(D) $\sin\left(10^3 t + 53^0\right)$

Q18. GATE ECE 2003

Open solution

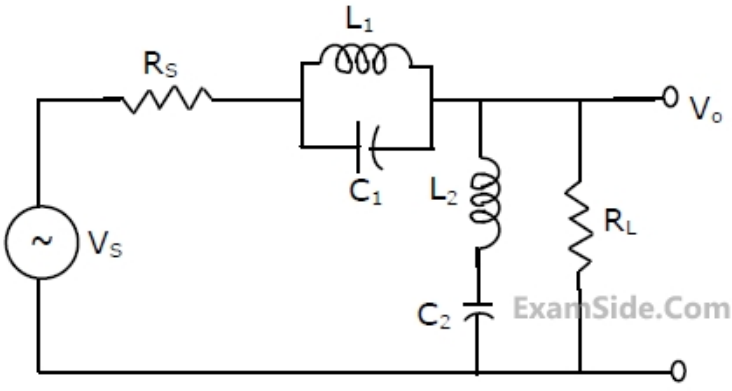
A series RLC circuit has a resonance frequency of 1 kHz and a quality factor Q = 100. If each R, L and C is doubled from its original value, the new Q of the circuit is

(A) 25 (B) 50 (C) 100 (D) 200

Q19. GATE ECE 2000

Open solution

The circuit of Fig. represents a



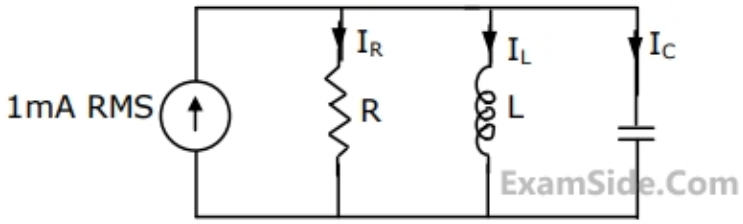
(A) Low pass filter
(B) High pass filter
(C) Band pass filter
(D) Band reject filter

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q20. GATE ECE 1998

Open solution

The parallel RLC circuit shown in figure is in resonance. In this circuit



(A) $|\mathrm{I}_R| < 1mA$

(B) $|\mathrm{I}_R + \mathrm{I}_L| > 1mA$

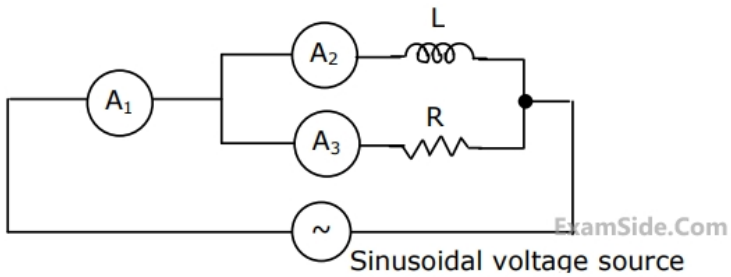
(C) $|\mathrm{I}_R + \mathrm{I}_C| < 1mA$

(D) $|\mathrm{I}_L + \mathrm{I}_C| > 1mA$

Q21. GATE ECE 1996

Open solution

In Fig., A_1, A_2 and A_3 are ideal ammeters. If A_2 and A_3 read 3 A and 4 A respectively, then A_1 should read



(A) $1\ A$

(B) $5\ A$

(C) $7\ A$

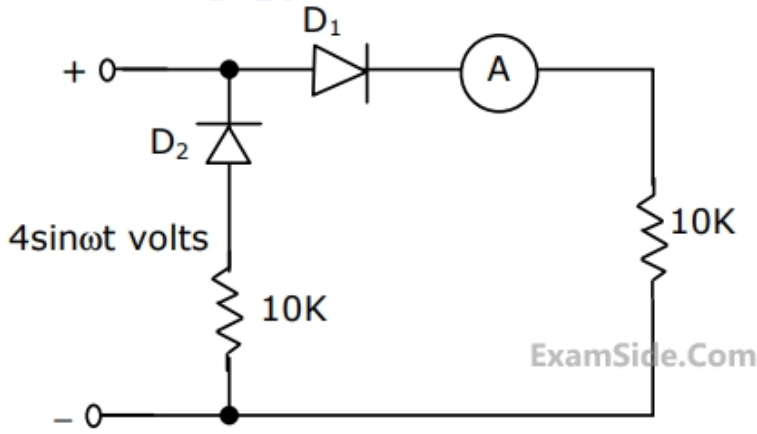
(D) None of the above

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q22. GATE ECE 1996

Open solution

In the circuit of Fig., assume that the diodes are ideal and the meter is an average indicating ammeter. The ammeter will read



(A) $0.4\sqrt{2}\ mA$ (B) $0.4\ mA$ (C) $\frac{0.8}{\pi}\ mA$ (D) $\frac{0.4}{\pi}\ mA$

Q23. GATE ECE 1995

Open solution

A DC voltage source is connected across a series R-L-C circuit. Under steady-state conditions, the applied DC voltage drops entirely across the

(A) R only (B) L only

(C) C only (D) R and L combination

Q24. GATE ECE 1995

Open solution

Consider a DC voltage source connected to a series R-C circuit. When the steady-state reaches, the ratio of the energy stored in the capacitor to the total energy supplied by the voltage source, is equal to

(A) 0.362 (B) 0.00 (C) 0.632 (D) 1.000

Q25. GATE ECE 1995

Open solution

The current $i(t)$ through a $10\text{-}\Omega$ resistor in series with an inductance is given by $i(t) = 3 + 4\sin(100t + 45^\circ) + 4\sin(300t + 60^\circ)\ A$.
The RMS value of the current and the power dissipated in the circuit are

(A) $\sqrt{41}\ A,\ 410\ W$, respectively (B) $\sqrt{35}\ A,\ 350\ W$, respectively

(C) $5\ A,\ 250\ W$, respectively (D) $11\ A,\ 1210\ W$,$ respectively

Q26. GATE ECE 1995

Open solution

A series R-L-C circuit has a Q of 100 and an impedance of $(100 + j0)\ \Omega$ at its resonant angular frequency of 10^7 rad/sec. The values of R and L are: $R =$ _____ ohms. $L =$ _____Henries.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q27. GATE ECE 1994

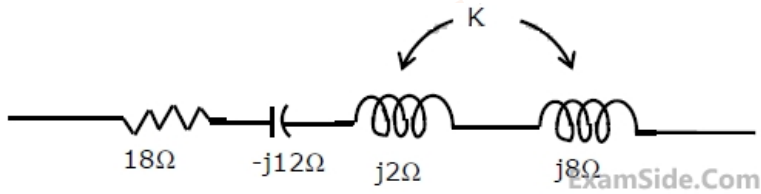
A series LCR circuit consisting of $R = 10\Omega$, $|X_L| = 20\Omega$ and $|X_C| = 20\Omega$, is connected across an a.c. supply of $200V$ rms. The rms voltage across the capacitor is

(A) $200\angle -90^\circ\, V$ (B) $200\angle +90^\circ\, V$ (C) $400\angle +90^\circ\, V$ (D) $400\angle -90^\circ\, V$

Open solution

Q28. GATE ECE 1993

In the series circuit shown in figure, for series resonance, the value of the coupling coefficient K will be



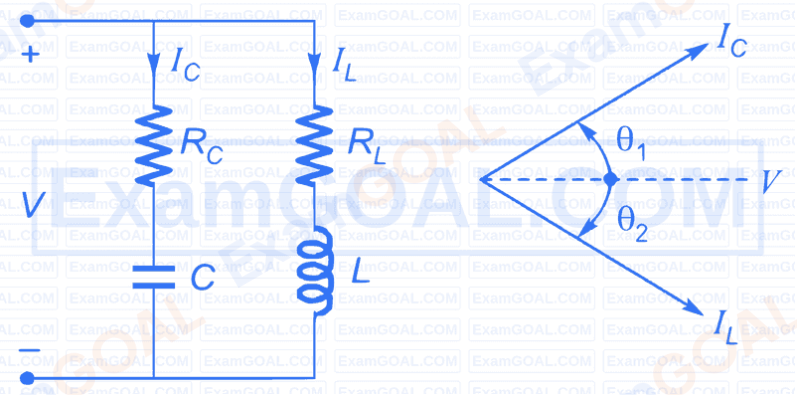
(A) 0.25 (B) 0.5 (C) 0.999 (D) 1.0

Open solution

Marks 2 38 questions

Q29. GATE ECE 2026

In the given circuit, $L = 1\mu\text{H}$ and $C = 1\mu\,\text{F}$. The phasor diagram for I_C and I_L is also shown. Assume that the phase $(\theta_1 + \theta_2)$ is 90° at a frequency of 159.15 kHz . Among the following options, what is the nearest integer value of $R_C \times R_L$?



(A) 0 (B) 1 (C) 2 (D) 10

Open solution

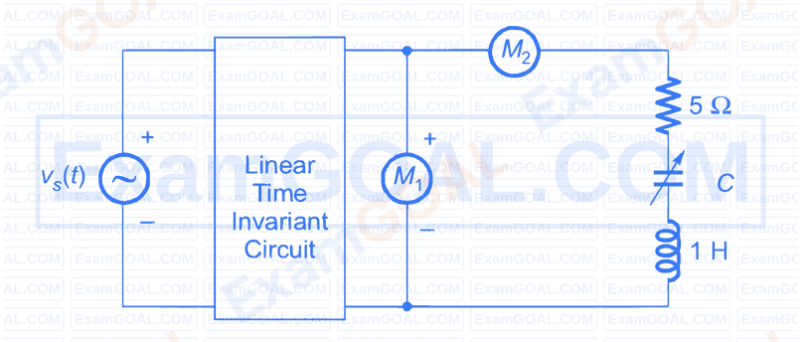
Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q30. GATE ECE 2025

Open solution

In the circuit below, M_1 is an ideal AC voltmeter and M_2 is an ideal AC ammeter. The source voltage (in Volts) is $v_s(t) = 100\cos(200t)$.

What should be the value of the variable capacitor C such that the RMS readings on M_1 and M_2 are 25 V and 5 A , respectively?



(A) $25\mu\,\mathrm{F}$

(B) $4\mu\,\mathrm{F}$

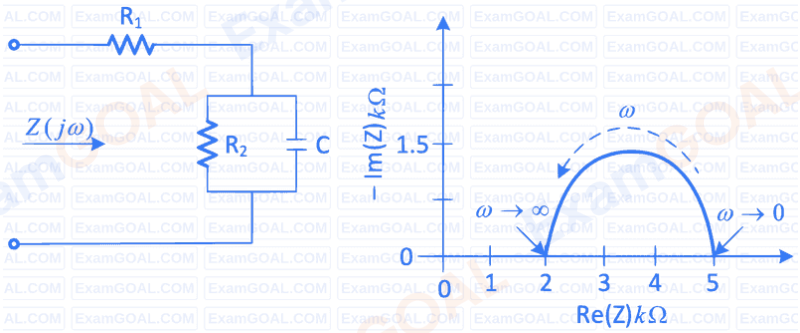
(C) $0.25\mu\,\mathrm{F}$

(D) Insufficient information to find C

Q31. GATE ECE 2022

Open solution

For the circuit shown, the locus of the impedance Z(jω) is plotted as ω increases from zero to infinity. The values of R_1 and R_2 are :



(A) R_1 = 2 kΩ, R_2 = 3 kΩ

(B) R_1 = 5 kΩ, R_2 = 2 kΩ

(C) R_1 = 5 kΩ, R_2 = 2.5 kΩ

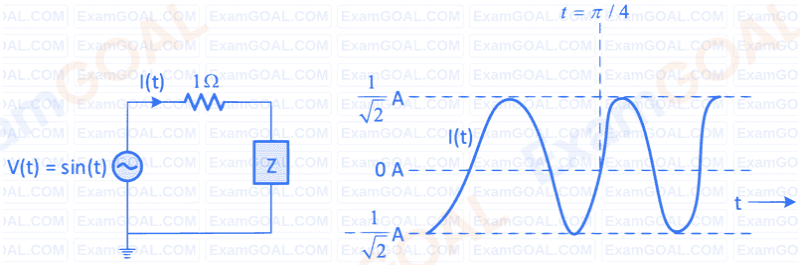
(D) R_1 = 2 kΩ, R_2 = 5 kΩ

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q32. GATE ECE 2022

Open solution

Consider the circuit shown in the figure with input V(t) in volts. The sinusoidal steady state current I(t) flowing through the circuit is shown graphically (where t is in seconds). The circuit element Z can be _____.



(A) a capacitor of 1 F

(B) an inductor of 1 H

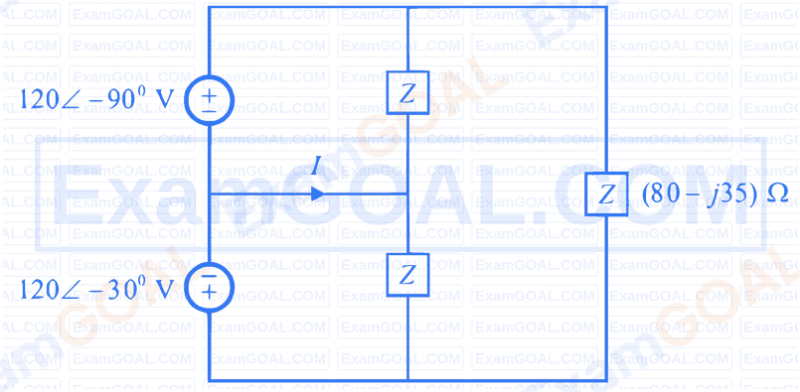
(C) a capacitor of $\sqrt{3}$F

(D) an inductor of $\sqrt{3}$H

Q33. GATE ECE 2020

Open solution

The current I in the given network is



(A) $2.38\angle -96.37^{\circ}\text{A}$.

(B) 0 A .

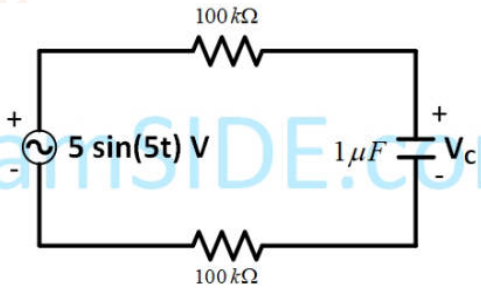
(C) $2.38\angle -23.63^{\circ}\text{A}$.

(D) $2.38\angle 143.63^{\circ}\text{A}$.

Q34. GATE ECE 2018

Open solution

For the circuit given in the figure, the voltage V_C (in volts) across the capacitor is :



(A) $1.25\sqrt{2}$ sin(5t - 0.25π)

(B) $1.25\sqrt{2}$ sin(5t - 0.125π)

(C) $2.5\sqrt{2}$ sin(5t - 0.25π)

(D) $2.5\sqrt{2}$ sin(5t - 0.125π)

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

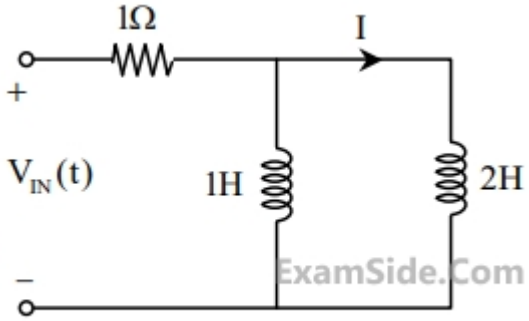
Q35. GATE ECE 2017 Set 1

Open solution

In the circuit shown the voltage $V_{IN}\ (t)$ is described by:

$$V_{IN}\ (t) = \begin{cases} 0, & for\ t < 0 \\ 15Volts, & for\ t \geq 0 \end{cases}$$

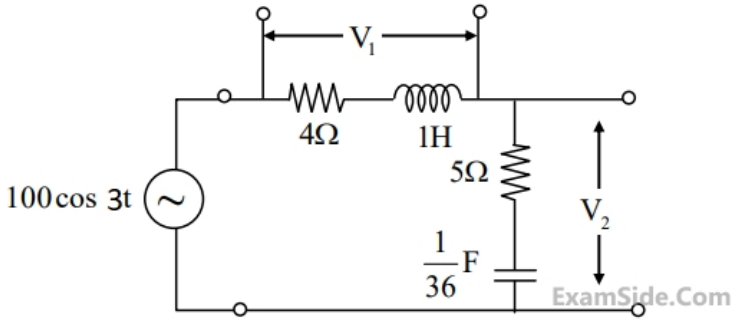
where 't' is in seconds. The time (in seconds) at which the current I in the circuit will reach the value 2 Ampere is _____ .



Q36. GATE ECE 2017 Set 1

Open solution

The figure shows an RLC circuit excited by the sinusoidal voltage $100cos(3t)$ Volts, where t is in seconds. The ratio $\frac{amplitude\ of\ V_2}{amplitude\ of\ V_1}$ is _____ .

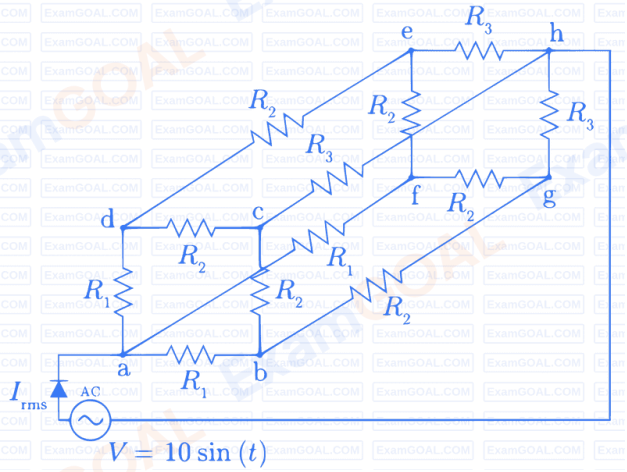


Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q37. GATE ECE 2016 Set 1

Open solution

An AC voltage source V = 10 sin(t) volts is applied to the following network. Assume that R_1 = 3 kΩ, R_2 = 6 kΩ and R_3 = 9 kΩ and that the diode is ideal.



RMS current I_{rms} (in mA) through the diode is _____.

Q38. GATE ECE 2016 Set 1

Open solution

A network consisting of a finite number of linear resistor (R), inductor (L), and capacitor (C) elements, connected all in series or all in parallel, is excited with a source of the form $\sum_{k=1}^{3} a_k \cos(k\omega_0 t)$, where $a_k \neq 0,\ \omega_0 \neq 0$.

The source has nonzero impedance. Which one of the following is a possible form of the output measured across a resistor in the network?

(A) $\sum_{k=1}^{3} b_k \cos(k\omega_0 t + \phi_k)$, where $b_k \neq a_k,\ \forall k$

(B) $\sum_{k=1}^{3} b_k \cos(k\omega_0 t + \phi_k)$, where $b_k \neq 0,\ \forall k$

(C) $\sum_{k=1}^{3} a_k \cos(k\omega_0 t + \phi_k)$

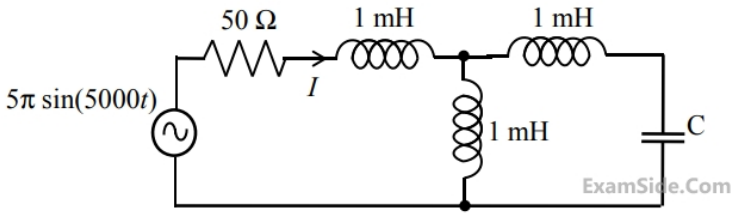
(D) $\sum_{k=1}^{3} a_k \cos(k\omega_0 t + \phi_k)$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q39. GATE ECE 2015 Set 3

Open solution

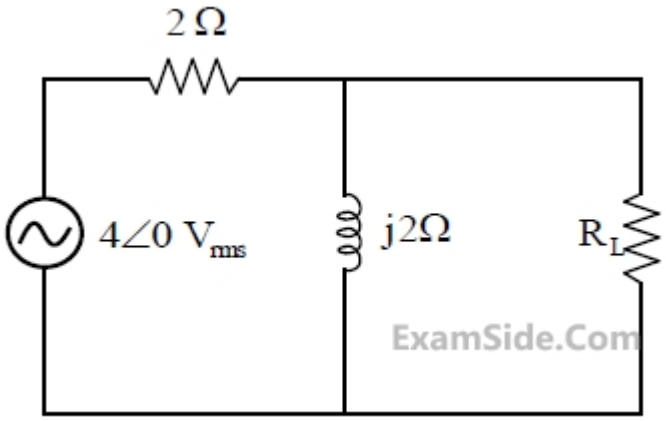
In the circuit shown, the current I flowing through the $50\,\Omega$ resistor will be zero if the value of capacitor **C** (in μF) is _____ .



Q40. GATE ECE 2015 Set 1

Open solution

In the given circuit, the maximum power (in Watts) that can be transferred to the load R_L is _____.



Q41. GATE ECE 2015 Set 1

Open solution

The damping ratio of a series RLC circuit can be expressed as

(A) $\frac{R^2C}{2L}$ (B) $\frac{2L}{R^2C}$ (C) $\frac{R}{2}\sqrt{\frac{C}{L}}$ (D) $\frac{2}{R}\sqrt{\frac{L}{C}}$

Q42. GATE ECE 2014 Set 1

Open solution

A 230 V rms source supplies power to two loads connected in parallel. The first load draws 10 kW at 0.8 leading power factor and the second one draws 10 kVA at 0.8 lagging power factor. The complex power delivered by the source is

(A) (18 + j 1.5) kVA (B) (18 - j 1.5) kVA (C) (20 + j 1.5) kVA (D) (20 - j 1.5) kVA

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q43. GATE ECE 2014 Set 1

A periodic variable x is shown in the figure as a function of time. The root - mean - square (rms) value of x is _____ .

Open solution



Q44. GATE ECE 2014 Set 1

In the circuit shown in the figure, the value of capacitor C (in mF) needed to have critically damped response i(t) is _____.

Open solution

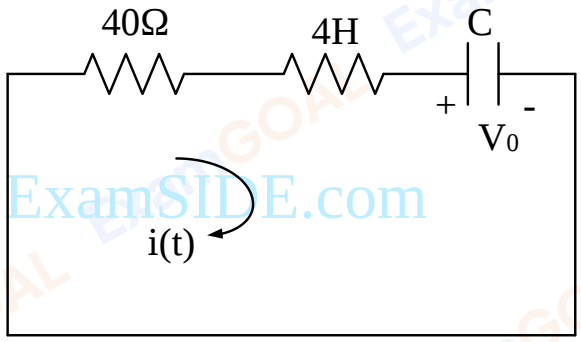


Fig.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

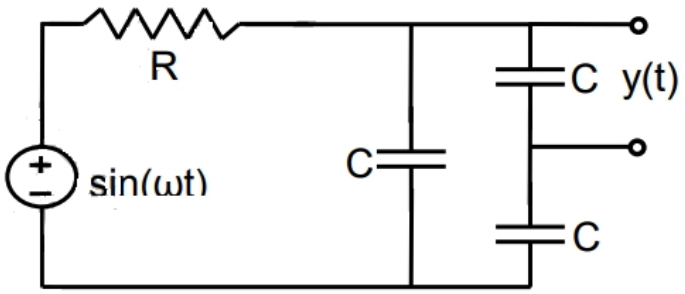
Q45. GATE ECE 2014 Set 4

Open solution

The steady state output of the circuit shown in the figure is given by

$$y(t) = \mathrm{A}(\omega) \sin(\omega t + \phi(\omega))$$

If the amplitude $|\mathrm{A}(\omega)| = 0.25$, then the frequency ω is



ExamSide.Com

Fig.

(A) $\frac{1}{\sqrt{3}RC}$ (B) $\frac{2}{\sqrt{3}RC}$ (C) $\frac{1}{RC}$ (D) $\frac{2}{RC}$

Q46. GATE ECE 2014 Set 2

Open solution

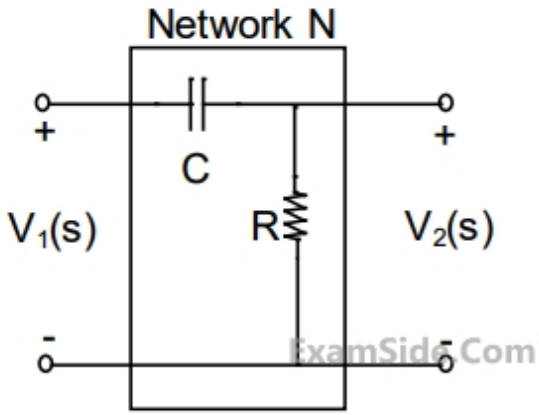
A series LCR circuit is operated at a frequency different from its resonant frequency. The operating frequency is such that the current leads the supply voltage. The magnitude of current is half the value at resonance. If the values of L, C and R are 1 H, 1 F and 1Ω , respectively, the operating angular frequency (in rad/s) is _____.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

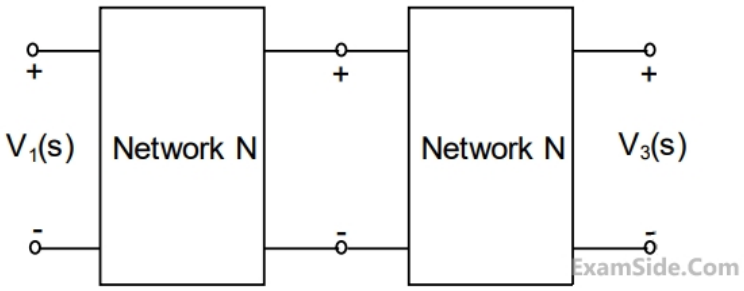
Q47. GATE ECE 2014 Set 3

Open solution

Consider the building block called 'Network N' shown in the figure.
Let $C = 100\,\mu F$ and $R = 10\,k\Omega$.



Two such blocks are connected in cascade, as shown in the figure.



The transfer function $\frac{V_3(s)}{V_1(s)}$ of the cascaded network is

(A) $\frac{s}{1+s}$

(B) $\frac{s^2}{1+3s+s^2}$

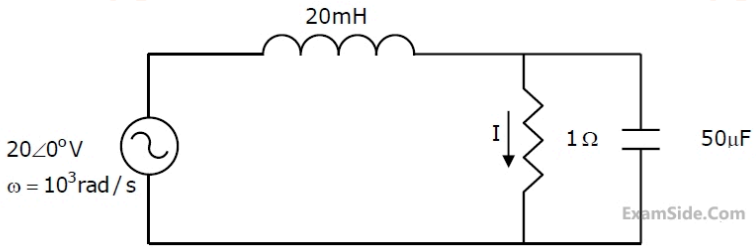
(C) $\left(\frac{s}{1+s}\right)^2$

(D) $\frac{s}{2+s}$

Q48. GATE ECE 2010

Open solution

The current I in the circuit shown is



(A) -j1 A

(B) j1 A

(C) 0 A

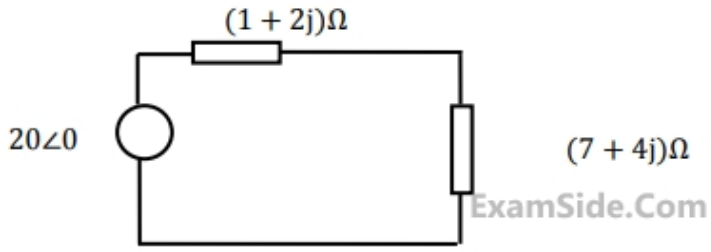
(D) 20 A

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q49. GATE ECE 2009

Open solution

An AC source of RMS voltage $20\ V$ with internal impedance $Z_s = (1+2j)\Omega$ feeds a load of impedance $Z_L = (7+4j)\Omega$ in the figure below. The reactive power consumed by the load is

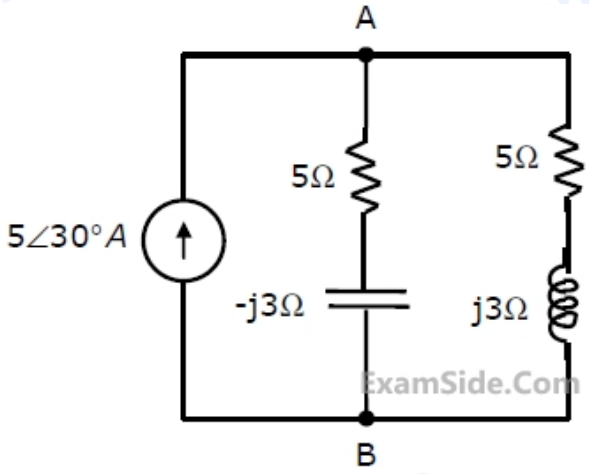


(A) 8 VAR (B) 16 VAR (C) 28 VAR (D) 32 VAR

Q50. GATE ECE 2007

Open solution

In the AC network shown in the figure, the phasor voltage V_{AB} (in Volts) is:



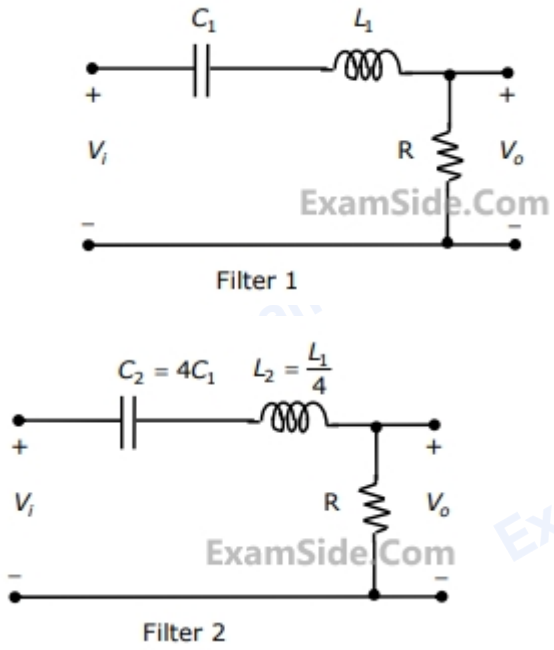
(A) 0 (B) $5\angle 30^\circ$ (C) $12.5\angle 30^\circ$ (D) $17\angle 30^\circ$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q51. GATE ECE 2007

Two series resonant filters are as shown in the figure. Let the 3-dB bandwidth of Filter 1 be B_1 and that of Filter 2 be B_2. The value of B_1/B_2 is

Open solution

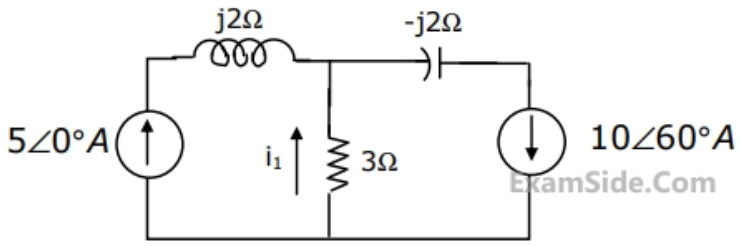


(A) 4 (B) 1 (C) $1/2$ (D) $1/4$

Q52. GATE ECE 2005

For the circuit in figure, the phase current I_1 is

Open solution



(A) $\frac{10\sqrt{3}}{2}\angle 90^0$ Amps (B) $\frac{10\sqrt{3}}{2}\angle -90^0$ Amps

(C) $5\angle 60^0$ Amps (D) $5\angle -60^0$ Amps

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q53. GATE ECE 2004

Open solution

Consider the following statements S1 and S2.

S_1: The β of a bipolar transistor reduces if the base width is increased.

S_2: The β of a bipolar transistor increases if the doping concentration in the base in increased

Which one of the following is correct?

(A) S_1 is FALSE and S_2 is TRUE

(B) Both S_1 and S_2 are TRUE

(C) S_1 is TRUE and S_2 is FALSE

(D) Both S_1 and S_2 are FALSE

Q54. GATE ECE 2004

Open solution

The transfer function $H(s) = \frac{V_0(s)}{V_i(s)}$ of an **R-L-C** circuit is given by

$H(s) = \frac{10^6}{s^2+20s+10^6}$

The Quality factor (Q-factore) of this circuit is

(A) 25

(B) 50

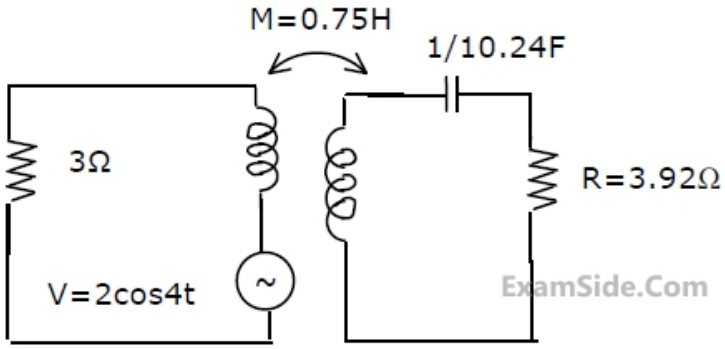
(C) 100

(D) 5000

Q55. GATE ECE 2003

Open solution

The current flowing through the resistance R in the circuit in figure has the form P cos 4t, where P is



(A) (0.18 + j 0.72)

(B) (0.46 + j 1.90)

(C) -(0.18 + j 1.90)

(D) -(0.192 + j 0.144)

Q56. GATE ECE 2003

Open solution

An input voltage $v(t) = 10\sqrt{2}\ \cos\ (t + 10^0) + 10\sqrt{5}\ \cos\ (2t + 10^0)\ V$ is applied to a series combination of resistance $R = 1\,\Omega$ and inductance $L = 1H$. the resulting steady - state current $i(t)$ in ampere is

(A) $10\cos\left(t + 55^0\right) + 10\ \cos\left(2t + 10^0 + \tan^{-1} 2\right)$

(B) $10\cos\left(t + 55^0\right) + 10\sqrt{\frac{3}{2}}\ \cos\left(2t + 55^0\right)$

(C) $10\cos\left(t - 35^0\right) + 10\cos\left(2t + 10^0 - \tan^{-1} 2\right)$

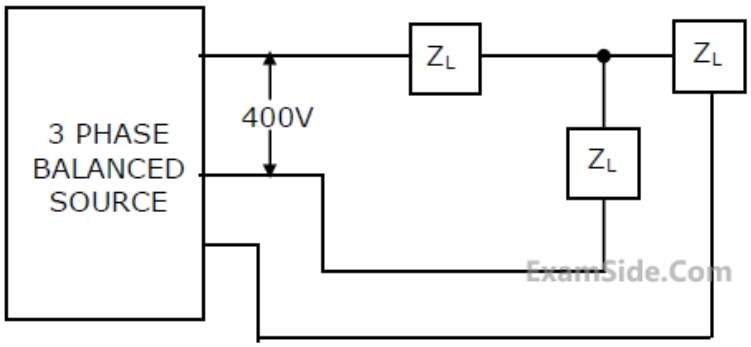
(D) $10\cos\left(t - 35^0\right) + 10\sqrt{\frac{3}{2}}\ \cos\left(2t - 35^0\right)$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q57. GATE ECE 2002

Open solution

If the 3-phase balanced source in Fig. delivers 1500 W at a leading power factor of 0.844, then the value of Z_L (in ohm) is approximately



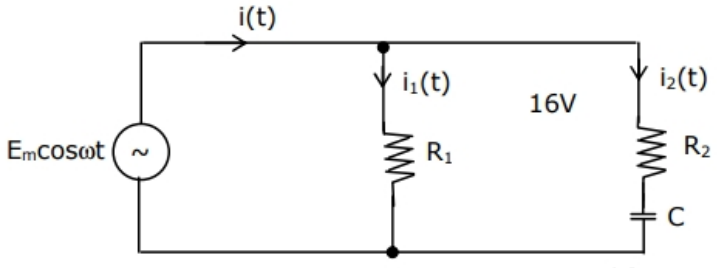
(A) $90\angle 32.44^\circ$ (B) $80\angle 32.44^\circ$ (C) $80\angle -32.44^\circ$ (D) $90\angle -32.44^\circ$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q58. GATE ECE 2001

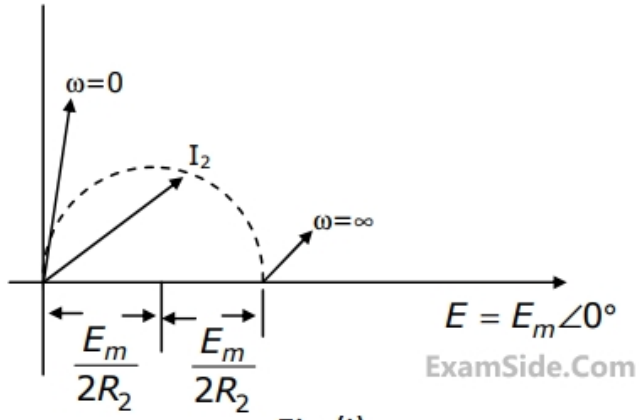
Open solution

When the angular frequency ω in Fig. is varied from 0 to ∞ the locus of the current phasor I_2 is given by



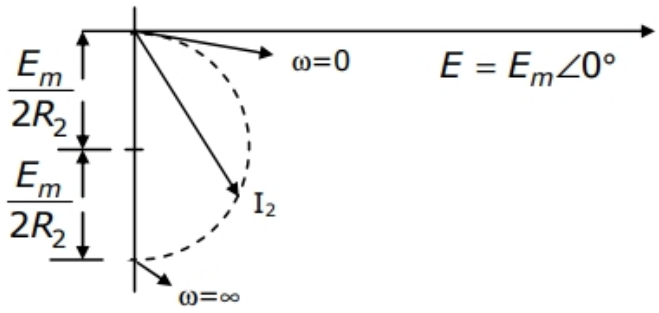
ExamSide.Com

Fig.



ExamSide.Com

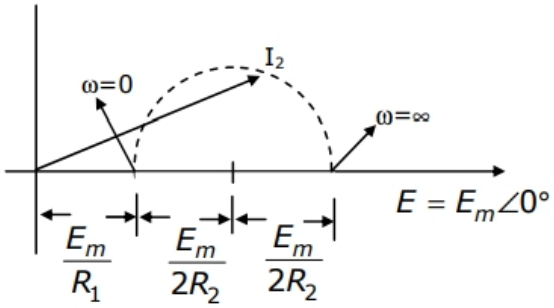
Fig.(i)



ExamSide.Com

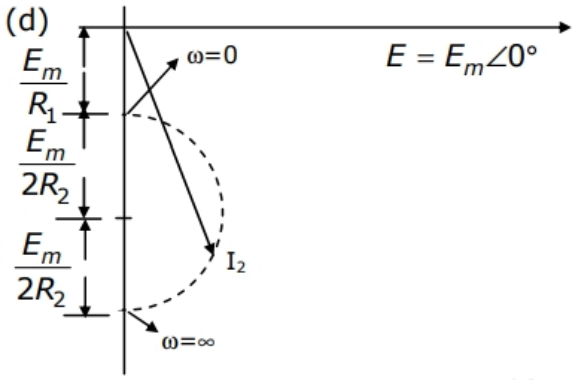
Fig.(ii)

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play



ExamSide.Com

Fig.(iii)



ExamSide.Com

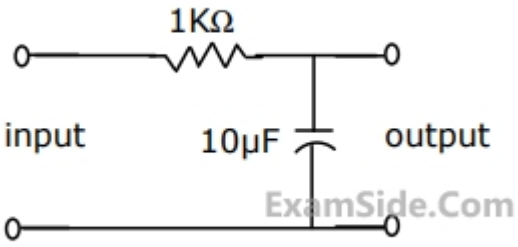
Fig.(iv)

(A) Fig. (i) (B) Fig. (ii) (C) Fig. (iii) (D) Fig. (iv)

Q59. GATE ECE 2000

In Fig., the steady state output voltage corresponding to the input voltage $(3 + 4\sin\ 100\,t)\ V$ is

Open solution



ExamSide.Com

(A) $3 + \frac{4}{\sqrt{2}}\sin\left(100\,t - \frac{\pi}{4}\right)\ V$

(B) $3 + 4\sqrt{2}\sin\left(100\,t - \frac{\pi}{4}\right)\ V$

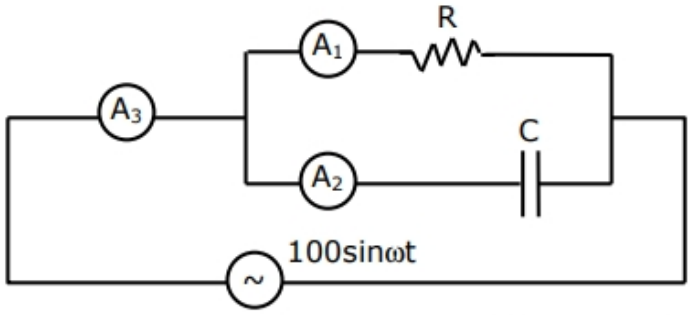
(C) $\frac{3}{2} + \frac{4}{\sqrt{2}}\sin\left(100\,t + \frac{\pi}{4}\right)\ V$

(D) $3 + 4\sin\left(100\,t - \frac{\pi}{4}\right)\ V$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q60. GATE ECE 1993

In figure, A_1, A_2 and A_3 are ideal ammeters. If A_1 reads 5 A, A_2, A_2 reads 12 A, then A_3 should read



ExamSide.Com

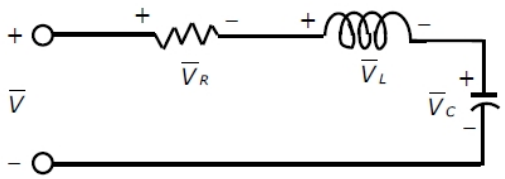
Fig.

Open solution

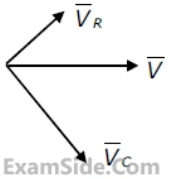
(A) 7A (B) 12A (C) 13A (D) 17A

Q61. GATE ECE 1992

For the series R-L-C circuit of figure(a), the partial phasor diagram at a certain frequency is shown in figure (b).The operating frequency of the circuit is:



(a)



ExamSide.Com

(b)

Open solution

(A) equal to the resonance frequency (B) less than the resonance frequency
(C) greater than the resonance frequency (D) not zero

Q62. GATE ECE 1991

In a series RLC high Q circuit, the current peaks at a frequency

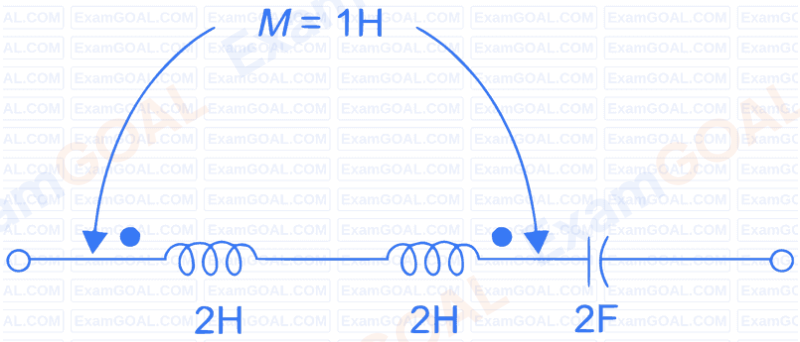
(A) equal to the resonant frequency (B) greater than the resonant frequency
(C) less than the resonant frequency (D) none of the above

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q63. GATE ECE 1990

The resonant frequency of this series circuit shown in figure is



Open solution

(A) $\frac{1}{4\pi\sqrt{3}}Hz$ (B) $\frac{1}{4\pi}Hz$ (C) $\frac{1}{2\pi\sqrt{10}}Hz$ (D) $\frac{1}{4\pi\sqrt{2}}Hz$

Q64. GATE ECE 1990

The transfer function of simple RC network functioning as a controller is

$$G_c(s)\ \frac{s+z_1}{s+p_1}$$

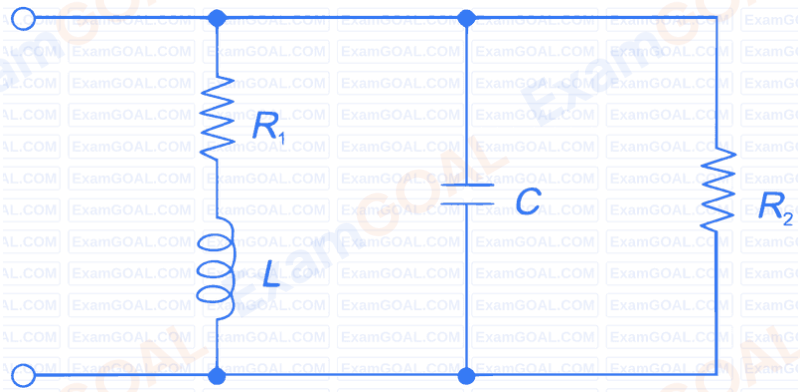
Open solution

The condition for the RC network to act as a phase lead controller is

(A) $p_1 < z_1$ (B) $p_1 = 0$ (C) $p_1 = z_1$ (D) $p_1 > z_1$

Q65. GATE ECE 1989

The half - power bandwidth of the resonant circuit of figure can be increased by



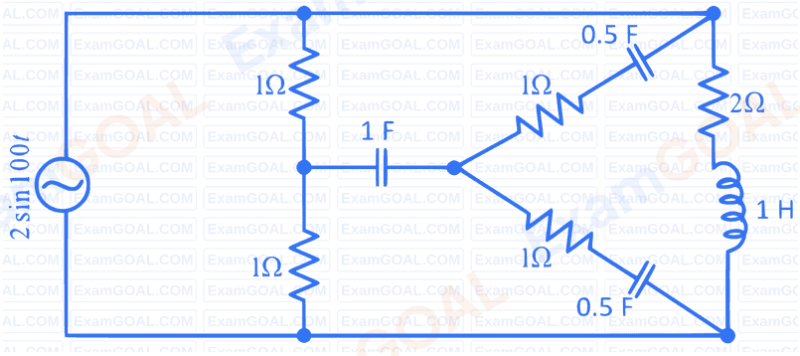
Open solution

(A) increasing R_1 (B) decreasing R_1 (C) increasing R_2 (D) decreasing R_2

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q66. GATE ECE 1987

The value of current through the 1F capacitor of figure is.

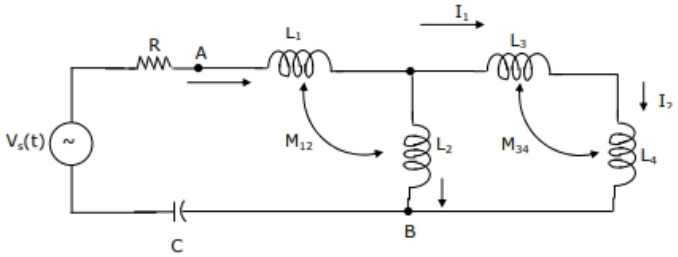


Open solution

Marks 5 8 questions

Q67. GATE ECE 2002

For network shown in Fig. $R = 1\,k\Omega$
$L_1 = 2\,H,\ L_2 = 5\,H,\ L_3 = 1H, L_4 = 4H$ and $C - 0.2\ \mu F$.. The mutual inductances are $M_{12} = 3\,H$ and $M_{34} = 2\,H$.
Determine
(a) the equivalent inductance for the combination of L_3 and L_4,
(b) the equivalent inductance across the points A and B in the network,
(c) the resonant frequency of the network.



ExamSide.Com

Fig.

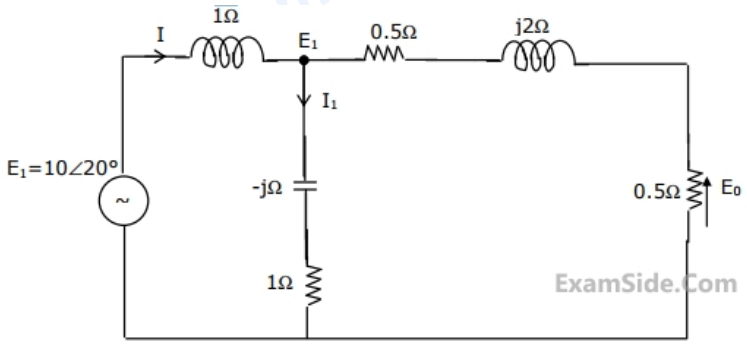
Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q68. GATE ECE 2001

Open solution

For the circuit shown in the figure, determine the phasors E_2, E_0, I and I_1.



Q69. GATE ECE 2000

Open solution

For the circuit in Fig. Which is in steady state,

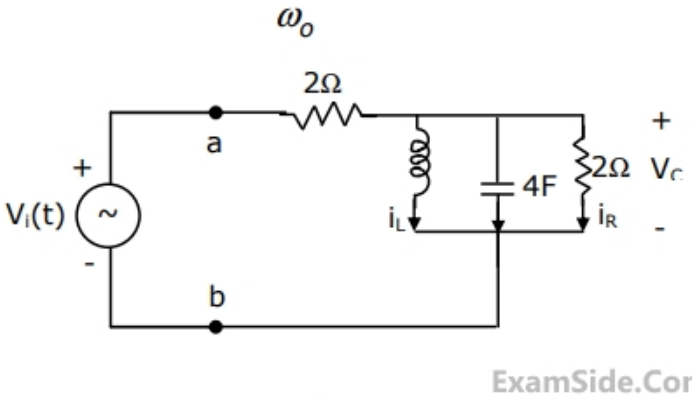


Fig.

(a)Find the frequency ω_0 at which the magnitude of the impedance across terminals a, b reaches maximum.

(b) Find the impedance across a, b at the frequency ω_0.

(c) If $v_i(t) = V \sin(\omega_0 t)$, find $i_L(t)$, $i_c(t)$, $i_R(t)$.

Q70. GATE ECE 1999

Open solution

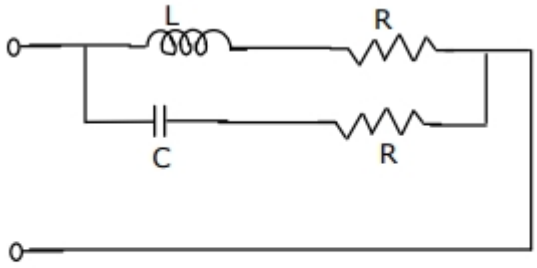
A coil with a quality factor (Q) of 10 is put in series with a capacitor C_1 of $10\ \mu F$, and the combination is found to draw maximum current when a sinusoidal voltage of frequency $50\ Hz$ is applied. A second capacitor C_2 is now in parallel with the circuit. What should be the capacitance of C_2 for combined circuit to act purely as a resistance for a sinusoidal excitation at a frequency of $100\ Hz$? Calculate the rms current drawn by the combined circuit at $100\ Hz$ if the applied voltage is $100V$ (rms).

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q71. GATE ECE 1998

Open solution

Determine the frequency of resonance and the resonant impedance of the parallel circuit shown in figure. What happens when $L = CR^2$?



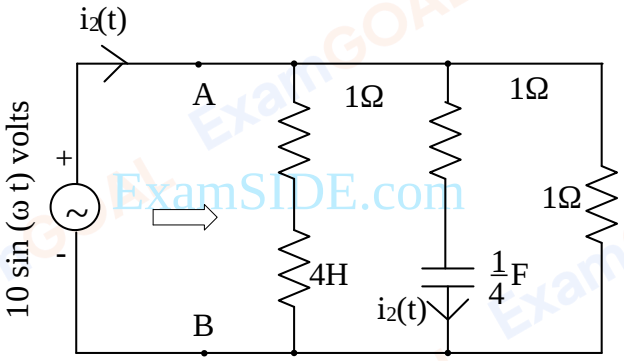
ExamSide.Com

Fig.

Q72. GATE ECE 1997

Open solution

In the circuit of Fig., all currents and voltage are sinusoids of frequency ω rad/sec.



ExamSIDE.com

Fig.

(a) Find the impedance to the right of $(A, \quad B)$ at $\omega = 0$ rad/sec and $\omega = \infty$ rad/sec.

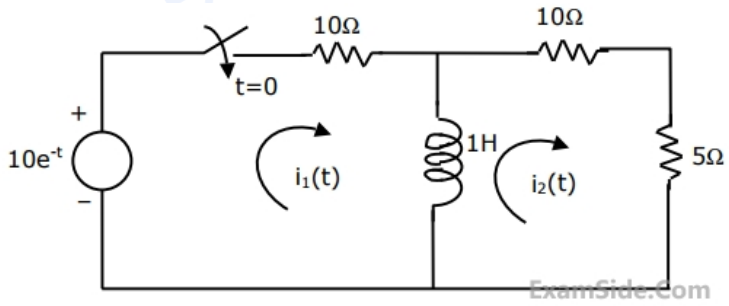
(b) If $\omega = \omega_0$ rad/sec and $i_1(t) = \mathrm{I} \sin(\omega_0 t)$ A, where I is positive, $\omega_0 \neq 0, \omega_0 \neq \infty$, then find I, ω_0 and $i_2(t)$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q73. GATE ECE 1994

Write down the mesh equation of the following network in terms of $i_1(t)$ and $i_2(t)$.Derive the differential equation for $i_1(t)$ from these and solve it.

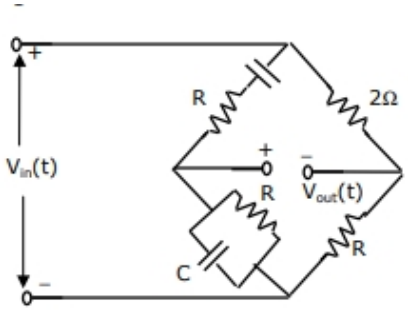
Open solution



Q74. GATE ECE 1994

Calculate the frequency at which zero- transmission is obtained from the Wien- bridge shown in Fig.

Open solution



ExamSide.Com

Fig.

Marks 8 1 questions

Q75. GATE ECE 1987

The impedance function of a parallel R, L, C circuit has poles located at $-3 \pm j4$ rad/sec. If the value of $L = 1H$, determine the following:
(a) the values of R and C
(b) the quality factor of the circuit

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Answer Key

Marks 1

1. **(72) (71.50 to 72.50)**	2. **A**	3. **D**	4. **2.80 to 2.85**	5. **649.5 to 651.5**
6. **0.2**	7. **1**	8. **0.32**	9. **B**	10. **0.5**
11. **100**	12. **25**	13. **D**	14. **C**	15. **B**
16. **A**	17. **A**	18. **B**	19. **D**	20. **B**
21. **B**	22. **D**	23. **C**	24. **B**	25. **C**
26. $R = 100\ \Omega$ $L = 1\ mH$	27. **D**	28. **A**		

Marks 2

29. **B**	30. **A**	31. **A**	32. **B**	33. **D**
34. **C**	35. **0.34**	36. **2.6**	37. **1**	38. **A**
39. **20**	40. **1.656**	41. **B**	42. **B**	43. **0.408**
44. **10**	45. **B**	46. **0.45**	47. **B**	48. **A**
49. **B**	50. **D**	51. **D**	52. **A**	53. **D**
54. **B**	55. **NULL**	56. **C**	57. **D**	58. **A**
59. **A**	60. **C**	61. **B**	62. **A**	63. **B**
64. **D**	65. **A, D**	66. **0**		

Marks 5

67. **(a)** $9\ H$ **(b)** $\frac{5}{7}\ H$ **(c)** $421\ Hz$	68. $E_2 = 8.77\ e^{j17.9^o}$ V $E_0 = 1.96\ e^{-j81.3^o}$ V $I = 6.202\ e^{-j9.9^o}$ A $I_1 = 6.2\ e^{-j27.1^o}$ A	69. **(a)** $W_0 = \frac{1}{4}\ r/s$ **(b)** 4Ω **(c)** $i_L(t) = -\frac{V}{2}cos\left(\frac{1}{4}t\right)$ A $i_C(t) = \frac{V}{2}cos\left(\frac{1}{4}t\right)$ A, $i_R(t) = \frac{2}{4}sin\left(\frac{1}{4}t\right)$ A	70. $C_2 = 3.33\ \mu\frac{x-\mu}{\sigma}F$ **RMS current** $= 3.14$ A	71. **Frequency of resonance** $= \frac{1}{2\pi\sqrt{LC}}$ **Resonant impedance** $= \frac{R^2+\frac{L}{C}}{2\ R}$
72. **(a)** $0.5\,\Omega,\ 0.5\ \Omega$ **(b)** $I = \frac{190}{17}$A, $W_0 = 1\ r/s,\quad i_2(t) = \frac{10}{\sqrt{17}}\sin\left(t + 76^0\right)$	73. **Try it.**	74. $\frac{1}{2\pi RC}\ Hz$		

Marks 8

75. **(a)** $R = \frac{25}{6}\Omega$, $C = \frac{1}{25}\ F$ **(b)** $Q = \frac{5}{6}$				

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

ExamGOAL
PYQ PDF

Your Exam. Our Responsibility.

From first chapter to exam day.

Practise with the questions your exam has already asked, then test yourself under real exam conditions.

PAST YEAR QUESTIONS

Every past paper, chapter-wise.

Previous year questions sorted the way you study, each with a full solution.

- Chapter-wise and year-wise
- Weightage and topic tags on every chapter
- Your progress on each one, at a glance

MOCK TESTS

Mock tests with a real exam feel.

Practise under the same pressure as the real paper, on your phone or on the big screen.

- Timed sections, just like exam day
- Mark questions for review and come back

ANALYSIS

Know your score. Know your rank.

A full report after every test, so you know exactly where you stand.

- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

NCERT Exemplar
With full solutions

Exam syllabus
Subject and chapter-wise

Bookmarks
Organised in groups

Your notes
On any question

ExamGOAL
LEARNING APP

ExamGOAL: Exam Preparation App

One app for every exam. Free trial for new users, on Android and the web.

GET IT ON Google Play

Scan to get the app

www.examgoal.com
Practice and mock tests

questions.examside.com
Free solutions

pdf.questions.examgoal.com
Free PYQ PDFs